Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Kooyman and Son Dairy

Physical address of dairy:

6787 E Jahant RD	Acampo	San Joaquin	95220
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1984

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X005-X250-XX14-XXXX

B. OPERATORS

Kooyman, Mike

Operator name: Kooyman, Mike

Telephone no.: (209) 367-4426 (Landline) (Cellular)

6787 E Jahant RD	Acampo	CA	95220
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Michael Ronald Kooyman Trust

Operator name: Michael Ronald Kooyman Trust

Telephone no.: (209) 367-4426 (Landline) (209) 608-4716 (Cellular)

6787 E Jahant RD	Acampo	CA	95220
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Michael Ronald Kooyman Trust

Legal owner name: Michael Ronald Kooyman Trust

Telephone no.: (209) 367-4426 (Landline) (209) 608-4716 (Cellular)

6787 E Jahant RD	Acampo	CA	95220
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	90	85	40	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	125	120	75	0
Average number	0	0	90	85	40	0
Avg live weight (lbs)	0	0	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,884.30 *tons per reporting period*

Total nitrogen from manure: 18,651.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 13,056.05 *lbs per reporting period*

Total phosphorus from manure: 2,955.04 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,804,500 *gallons*

Total nitrogen generated: 15,406.00 *lbs*

Total phosphorus generated: 1,784.58 *lbs*

Total potassium generated: 57,642.49 *lbs*

Total salt generated: 368,754.28 *lbs*

	7,804,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,804,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Wells	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	15	15	2	process wastewater	X005-X250-XX13-XXXX
2	10	10	2	process wastewater	X005-X250-XX13-XXXX
3	14	14	2	process wastewater	X005-X250-XX14-XXXX
4	12	12	1	process wastewater	X005-X250-XX14-XXXX
5	20	20	1	process wastewater	X005-X250-XX14-XXXX
Totals for areas that were used for application	71	71	8		
Totals for areas that were not used for application					
Land application area totals	71	71	8		

B. CROPS AND HARVESTS

1

Field name: 1

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	246.00 *ton*	As-is		70.0	3,390.00	790.00	6,300.00		9.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.40	111.19	25.91	206.64	894.46

1

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	414.00 *ton*	As-is		67.2	3,380.00	500.00	3,280.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.60	186.58	27.60	181.06	814.75

2

Field name: 2

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	164.00 *ton*	As-is		70.0	3,390.00	790.00	6,300.00		9.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.40	111.19	25.91	206.64	894.46

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	276.00 *ton*	As-is		67.2	3,380.00	500.00	3,280.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.60	186.58	27.60	181.06	814.75

3

Field name: 3

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	229.60 *ton*	As-is		70.0	3,390.00	790.00	6,300.00		9.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.40	111.19	25.91	206.64	894.46

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	386.40 *ton*	As-is		67.2	3,380.00	500.00	3,280.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.60	186.58	27.60	181.06	814.75

4

Field name: 4

10/20/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 12 Plant date: 10/20/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	102.00 *ton*	As-is		17.2	19,600.00	2,850.00	32,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.50	333.20	48.45	559.30	1,534.28

5

Field name: 5

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	330.00 *ton*	As-is		70.0	3,540.00	840.00	6,900.00		8.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	116.82	27.72	227.70	853.38

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/03/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.74	4.58	137.77	1,629.54	462,000.00 *gal*
Application event totals		72.74	4.58	137.77	1,629.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.54	4.25	127.93	1,513.15	429,000.00 *gal*
Application event totals		67.54	4.25	127.93	1,513.15	

1 - 06/05/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.02	8.34	309.06	1,974.33	627,000.00 *gal*
Irrigation Wells	Ground water	10.12	0.00	0.00	275.51	1,368,000.00 *gal*
Application event totals		93.14	8.34	309.06	2,249.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	14.38	0.00	0.00	391.51	1,944,000.00 *gal*
Application event totals		14.38	0.00	0.00	391.51	

1 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.32	0.00	0.00	362.51	1,800,000.00 *gal*
Application event totals		13.32	0.00	0.00	362.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.65	15.00	477.88	1,646.44	627,000.00 *gal*
Irrigation Wells	Ground water	10.12	0.00	0.00	275.51	1,368,000.00 *gal*
Application event totals		69.77	15.00	477.88	1,921.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.79	0.00	0.00	348.01	1,728,000.00 *gal*
Application event totals		12.79	0.00	0.00	348.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.25	0.00	0.00	333.51	1,656,000.00 *gal*
Application event totals		12.25	0.00	0.00	333.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.72	0.00	0.00	319.01	1,584,000.00 *gal*
Application event totals		11.72	0.00	0.00	319.01	

2 - 10/03/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.04	4.66	140.23	1,658.64	313,500.00 *gal*
Application event totals		74.04	4.66	140.23	1,658.64	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.14	4.41	132.85	1,571.35	297,000.00 *gal*
Application event totals		70.14	4.41	132.85	1,571.35	

2 - 06/05/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.20	8.56	317.19	2,026.28	429,000.00 *gal*
Irrigation Wells	Ground water	10.39	0.00	0.00	282.76	936,000.00 *gal*
Application event totals		95.59	8.56	317.19	2,309.04	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	14.38	0.00	0.00	391.51	1,296,000.00 *gal*
Application event totals		14.38	0.00	0.00	391.51	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.58	0.00	0.00	369.76	1,224,000.00 *gal*
Application event totals		13.58	0.00	0.00	369.76	

2 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.51	14.21	452.73	1,559.78	396,000.00 *gal*
Irrigation Wells	Ground water	9.59	0.00	0.00	261.00	864,000.00 *gal*
Application event totals		66.10	14.21	452.73	1,820.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.19	0.00	0.00	358.88	1,188,000.00 *gal*
Application event totals		13.19	0.00	0.00	358.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.79	0.00	0.00	348.01	1,152,000.00 *gal*
Application event totals		12.79	0.00	0.00	348.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.19	0.00	0.00	304.51	1,008,000.00 *gal*
Application event totals		11.19	0.00	0.00	304.51	

3 - 10/03/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.37	4.55	137.06	1,621.23	429,000.00 *gal*
Application event totals		72.37	4.55	137.06	1,621.23	

3 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.58	4.38	131.79	1,558.88	412,500.00 *gal*
Application event totals		69.58	4.38	131.79	1,558.88	

3 - 06/05/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.27	8.46	313.70	2,004.02	594,000.00 *gal*
Irrigation Wells	Ground water	10.27	0.00	0.00	279.65	1,296,000.00 *gal*
Application event totals		94.54	8.46	313.70	2,283.66	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.70	0.00	0.00	372.86	1,728,000.00 *gal*
Application event totals		13.70	0.00	0.00	372.86	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.13	0.00	0.00	357.33	1,656,000.00 *gal*
Application event totals		13.13	0.00	0.00	357.33	

3 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.18	14.38	458.12	1,578.35	561,000.00 *gal*
Irrigation Wells	Ground water	9.70	0.00	0.00	264.11	1,224,000.00 *gal*
Application event totals		66.89	14.38	458.12	1,842.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.13	0.00	0.00	357.33	1,656,000.00 *gal*
Application event totals		13.13	0.00	0.00	357.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.99	0.00	0.00	326.26	1,512,000.00 *gal*
Application event totals		11.99	0.00	0.00	326.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.99	0.00	0.00	326.26	1,512,000.00 *gal*
Application event totals		11.99	0.00	0.00	326.26	

4 - 10/20/2018: Alfalfa, hay

Field name: 4

Crop: Alfalfa, hay

Plant date: 10/20/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.65	0.00	0.00	344.38	1,368,000.00 *gal*
Application event totals		12.65	0.00	0.00	344.38	

4 - 10/20/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.32	0.00	0.00	362.51	1,440,000.00 *gal*
Application event totals		13.32	0.00	0.00	362.51	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.65	0.00	0.00	371.57	1,476,000.00 *gal*
Application event totals		13.65	0.00	0.00	371.57	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.32	0.00	0.00	362.51	1,440,000.00 *gal*
Application event totals		13.32	0.00	0.00	362.51	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.86	14.80	471.60	1,624.77	495,000.00 *gal*
Irrigation Wells	Ground water	9.99	0.00	0.00	271.88	1,080,000.00 *gal*
Application event totals		68.85	14.80	471.60	1,896.65	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.32	0.00	0.00	362.51	1,440,000.00 *gal*
Application event totals		13.32	0.00	0.00	362.51	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.65	0.00	0.00	371.57	1,476,000.00 *gal*
Application event totals		13.65	0.00	0.00	371.57	

4 - 10/20/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.86	14.80	471.60	1,624.77	495,000.00 *gal*
Irrigation Wells	Ground water	9.99	0.00	0.00	271.88	1,080,000.00 *gal*
Application event totals		68.85	14.80	471.60	1,896.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.65	0.00	0.00	344.38	1,368,000.00 *gal*
Application event totals		12.65	0.00	0.00	344.38	

5 - 10/03/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.04	4.66	140.23	1,658.64	627,000.00 *gal*
Application event totals		74.04	4.66	140.23	1,658.64	

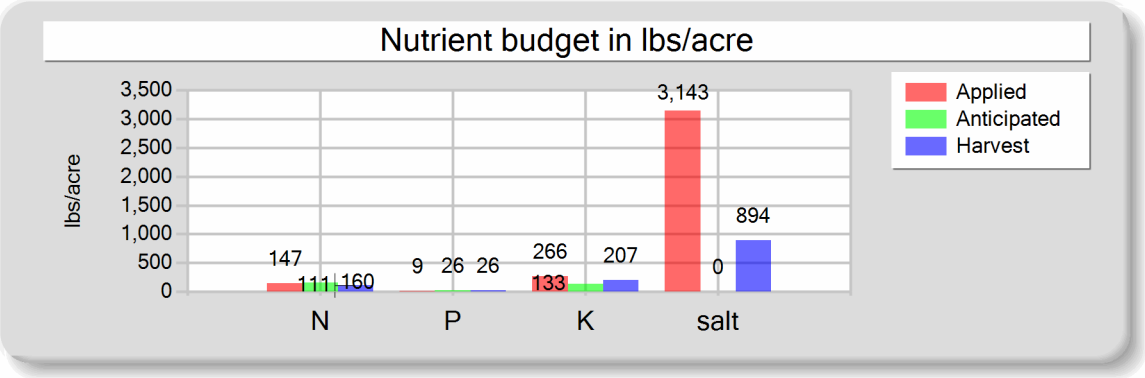
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.09	4.53	136.54	1,615.00	610,500.00 *gal*
Application event totals		72.09	4.53	136.54	1,615.00	

B. NUTRIENT BUDGET

1 - 10/03/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	140.28	8.82	265.69	3,142.69
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	147.28	8.82	265.69	3,142.69
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	111.19	25.91	206.64	894.46
Nutrient balance	36.09	-17.09	59.05	2,248.24
Applied to removed ratio	1.32	0.34	1.29	3.51

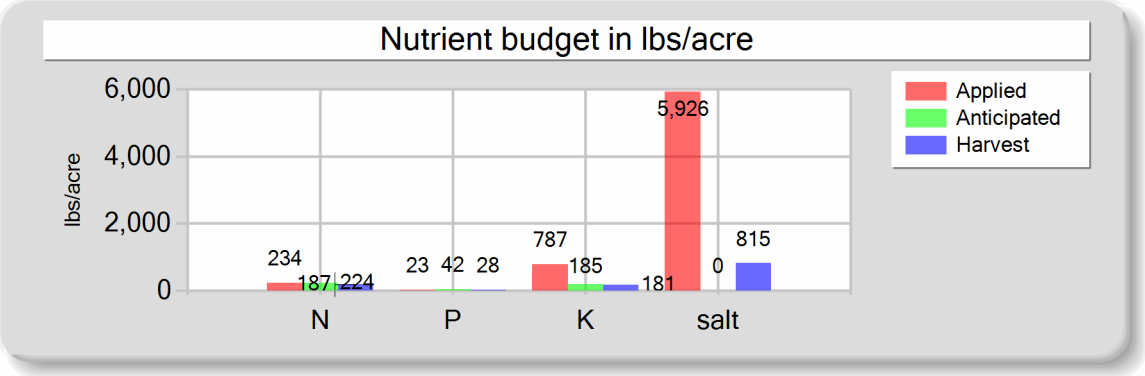
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
891,000.00	*gallons*
32.81	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/05/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	142.67	23.34	786.94	3,620.76
Fresh water	84.71	0.00	0.00	2,305.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.37	23.34	786.94	5,926.31
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	186.58	27.60	181.06	814.75
Nutrient balance	47.80	-4.26	605.88	5,111.55
Applied to removed ratio	1.26	0.85	4.35	7.27

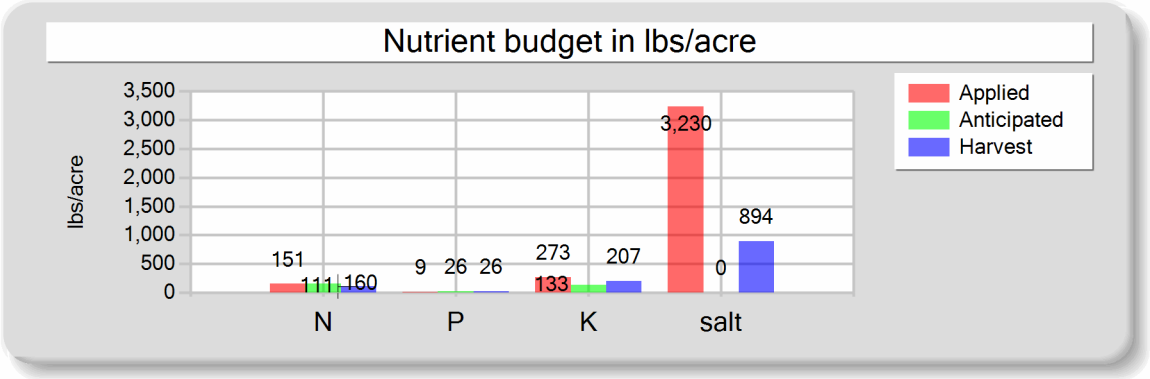
Fresh water applied	
11,448,000.00	*gallons*
421.59	*acre-inches*
28.11	*inches/acre*

Process wastewater applied	
1,254,000.00	*gallons*
46.18	*acre-inches*
3.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/03/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	144.18	9.07	273.07	3,229.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.18	9.07	273.07	3,229.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	111.19	25.91	206.64	894.46
Nutrient balance	39.99	-16.84	66.43	2,335.53
Applied to removed ratio	1.36	0.35	1.32	3.61

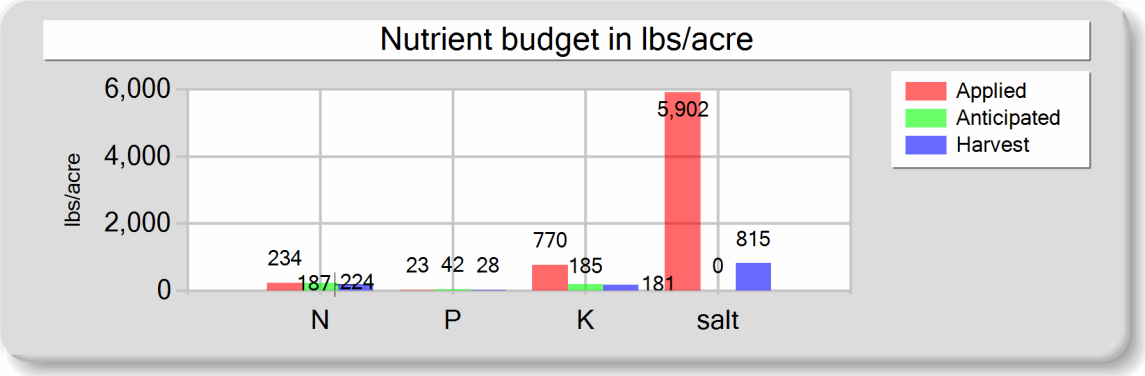
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
610,500.00 *gallons*
22.48 *acre-inches*
2.25 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 06/05/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.71	22.77	769.92	3,586.06
Fresh water	85.11	0.00	0.00	2,316.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.82	22.77	769.92	5,902.48
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	186.58	27.60	181.06	814.75
Nutrient balance	47.24	-4.83	588.87	5,087.73
Applied to removed ratio	1.25	0.82	4.25	7.24

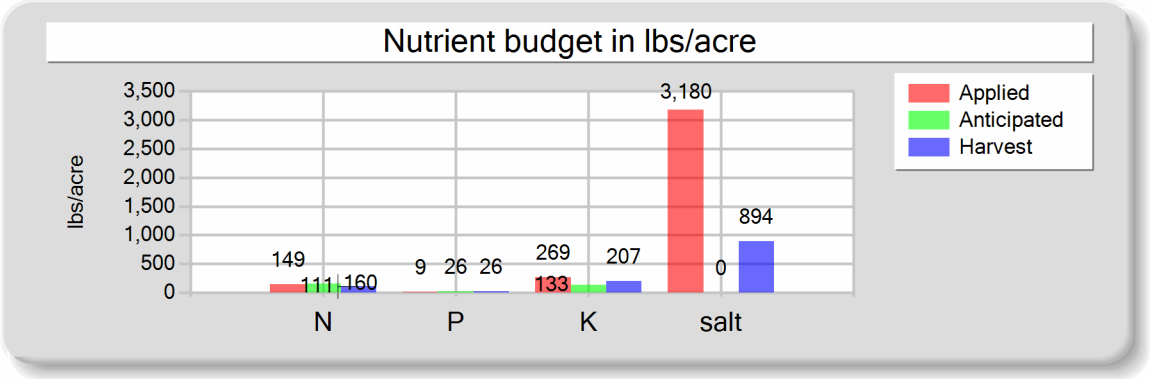
Fresh water applied	
7,668,000.00	*gallons*
282.39	*acre-inches*
28.24	*inches/acre*

Process wastewater applied	
825,000.00	*gallons*
30.38	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/03/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.95	8.93	268.85	3,180.11
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	148.95	8.93	268.85	3,180.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	111.19	25.91	206.64	894.46
Nutrient balance	37.76	-16.98	62.21	2,285.65
Applied to removed ratio	1.34	0.34	1.30	3.56

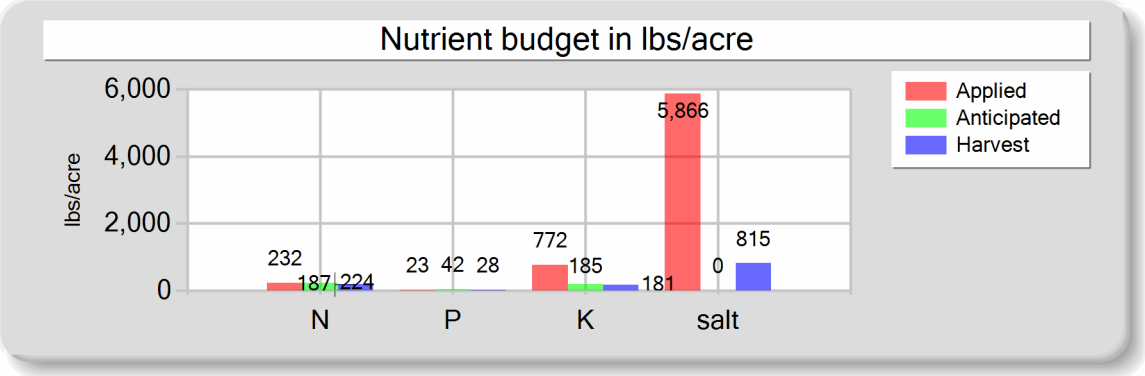
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
841,500.00 *gallons*
30.99 *acre-inches*
2.21 *inches/acre*

Total harvests for the crop
1 *harvests*

3 - 06/05/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.45	22.84	771.83	3,582.37
Fresh water	83.91	0.00	0.00	2,283.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.36	22.84	771.83	5,866.16
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	186.58	27.60	181.06	814.75
Nutrient balance	45.78	-4.76	590.77	5,051.41
Applied to removed ratio	1.25	0.83	4.26	7.20

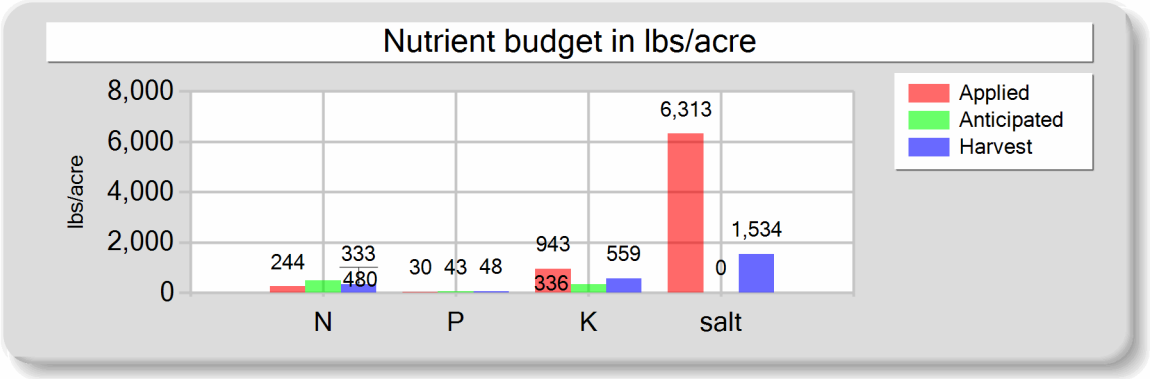
Fresh water applied	
10,584,000.00	*gallons*
389.77	*acre-inches*
27.84	*inches/acre*

Process wastewater applied	
1,155,000.00	*gallons*
42.53	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/20/2018: Alfalfa, hay

Field name: 4 Crop: Alfalfa, hay Plant date: 10/20/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	117.73	29.60	943.19	3,249.54
Fresh water	112.54	0.00	0.00	3,063.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	244.27	29.60	943.19	6,312.73
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	333.20	48.45	559.30	1,534.28
Nutrient balance	-88.93	-18.85	383.89	4,778.44
Applied to removed ratio	0.73	0.61	1.69	4.11

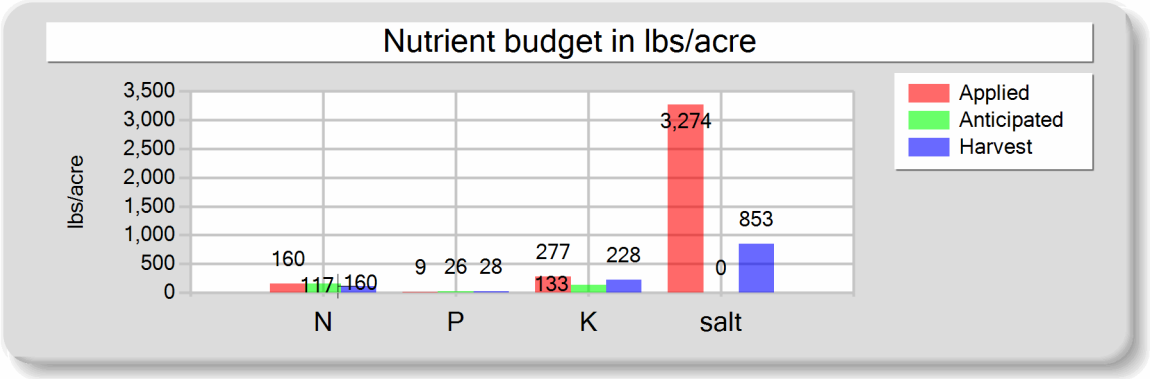
Fresh water applied	
12,168,000.00	*gallons*
448.11	*acre-inches*
37.34	*inches/acre*

Process wastewater applied	
990,000.00	*gallons*
36.46	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/03/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	146.13	9.19	276.76	3,273.64
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	160.13	9.19	276.76	3,273.64
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	116.82	27.72	227.70	853.38
Nutrient balance	43.31	-18.53	49.06	2,420.26
Applied to removed ratio	1.37	0.33	1.22	3.84

Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,237,500.00	*gallons*
45.57	*acre-inches*
2.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,100.00	2,500.00	13,300.00	8,600.00	5,800.00	2,000.00	1,500.00	0.00		87.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	4,610.00	36,600.00							31.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

1QWW

Sample and source description: 1QWW

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	283.00	202.00	0.00	0.00	17.80	536.00	89.30	50.30	131.00	2,120.00	0.00	179.00	352.00	5,270.00	6,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	1.00	0.10	1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	238.00	157.00	0.00	0.00	23.90	886.00								4,990.00	5,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	171.00	81.20	0.00	0.00	43.00	1,370.00								6,110.00	4,720
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	227.00	58.80	0.00	0.00	56.20	1,720.00								7,860.00	5,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Irrigation Wells

Irrigation Wells

IW #1

Sample description: IW #1

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.90	0.00	12.70								549.00	350
DL	1.00	0.50	0.10								10.00	10

IW #11

Sample description: IW #11

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.30	0.00	2.30								247.00	160
DL	1.00	0.50	0.10								10.00	10

IW #3

Sample description: IW #3

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.30	0.00	13.00								549.00	362
DL	1.00	0.50	0.10								10.00	10

Irrigation Wells

IW #4

Sample description: IW #4

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.06	0.00	1.90								218.00	145
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

1

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

2

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

3

3

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

4

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			73.00				
DL			4.00				

5

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			110.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 10/03/2022: Oats, silage-soft dough

1 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	790.00	6,300.00		9.09
DL	100.00	100.00	100.00		0.01

1 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	500.00	3,280.00		4.50
DL	100.00	100.00	100.00		0.01

2 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	790.00	6,300.00		9.09
DL	100.00	100.00	100.00		0.01

2 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	500.00	3,280.00		4.50
DL	100.00	100.00	100.00		0.01

3 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	790.00	6,300.00		9.09
DL	100.00	100.00	100.00		0.01

3 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	500.00	3,280.00		4.50
DL	100.00	100.00	100.00		0.01

4 - 10/20/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	2,850.00	32,900.00		10.90
DL	100.00	100.00	100.00		0.01

5 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,540.00	840.00	6,900.00		8.62
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

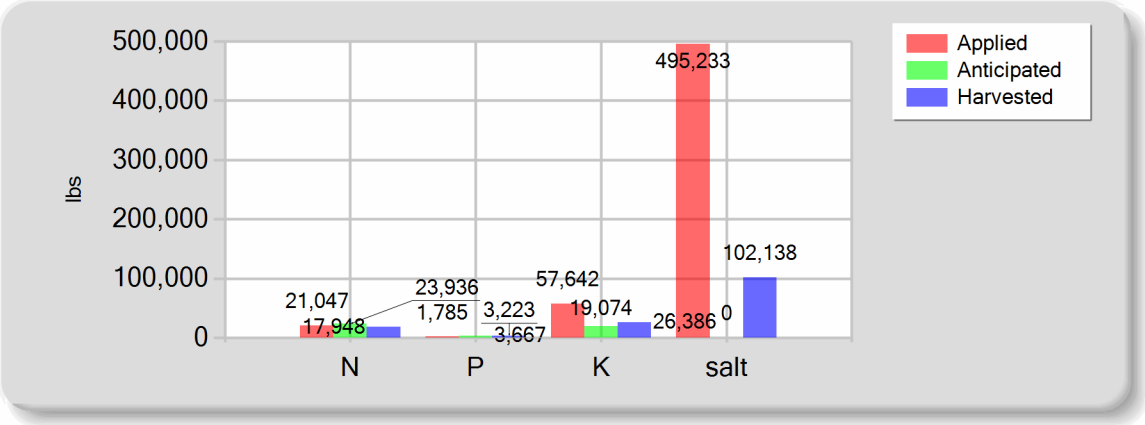
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

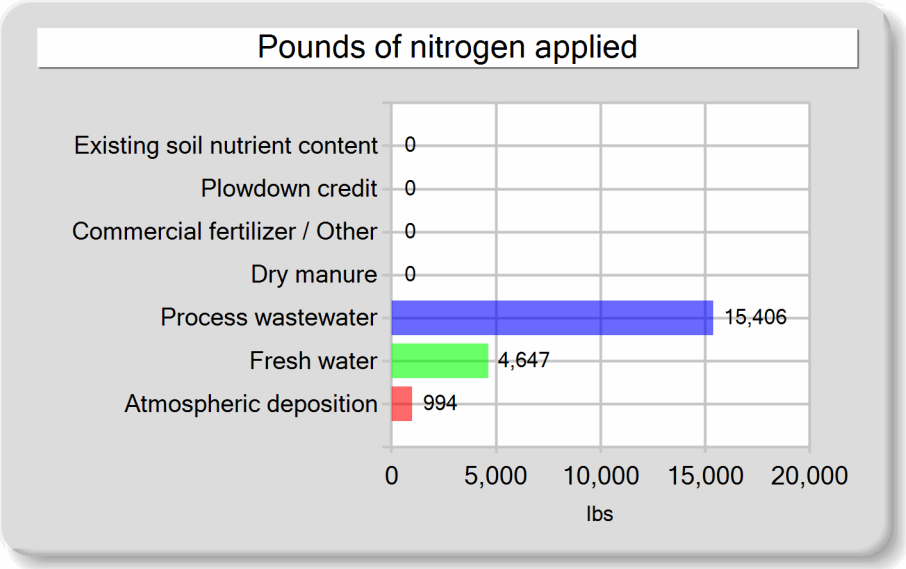
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

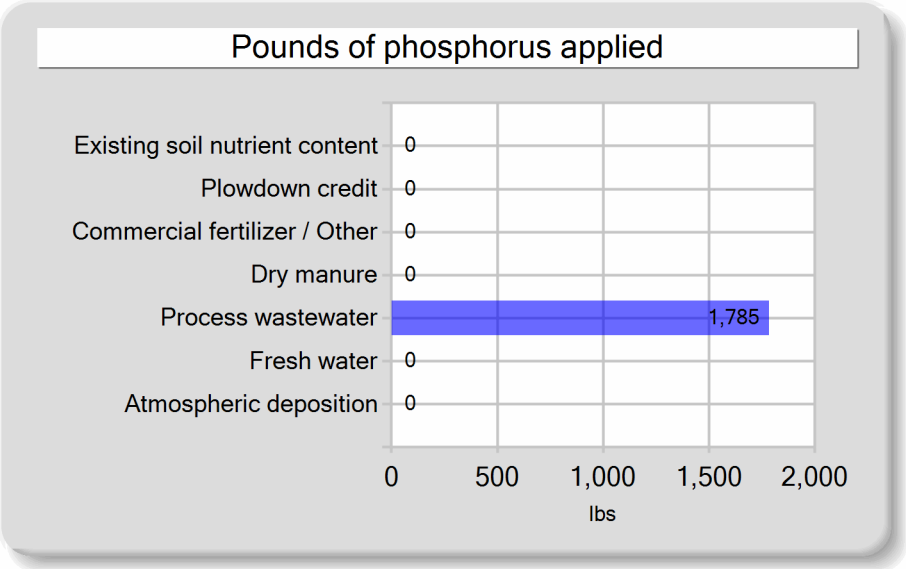
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	15,406.00	1,784.58	57,642.49	368,754.28
Fresh water	4,646.87	0.00	0.00	126,478.62
Atmospheric deposition	994.00	0.00	0.00	0.00
Total nutrients applied	21,046.87	1,784.58	57,642.49	495,232.91
Anticipated crop nutrient removal	23,936.00	3,666.80	19,074.40	0.00
Actual crop nutrient removal	17,947.75	3,222.77	26,385.74	102,138.12
Nutrient balance	3,099.12	-1,438.19	31,256.74	393,094.79
Applied to removed ratio	1.17	0.55	2.18	4.85

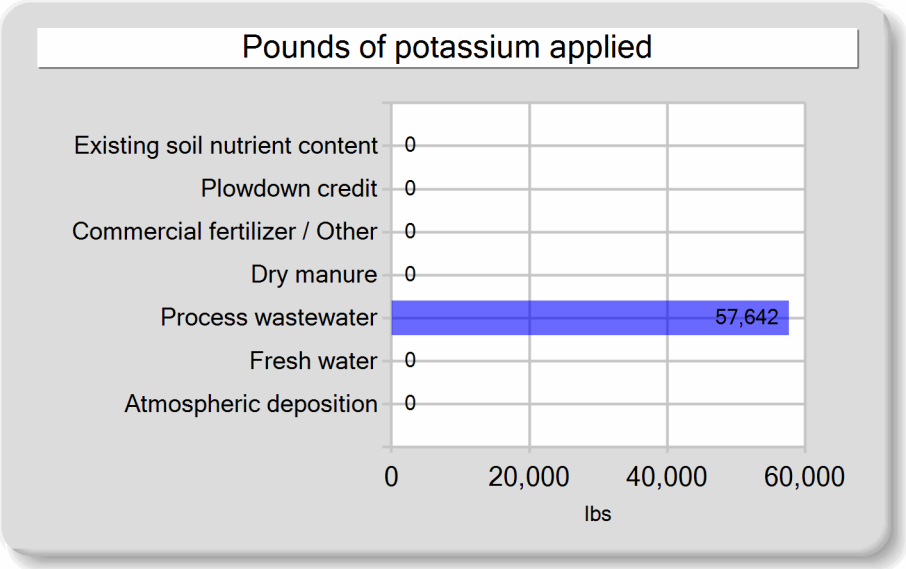
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

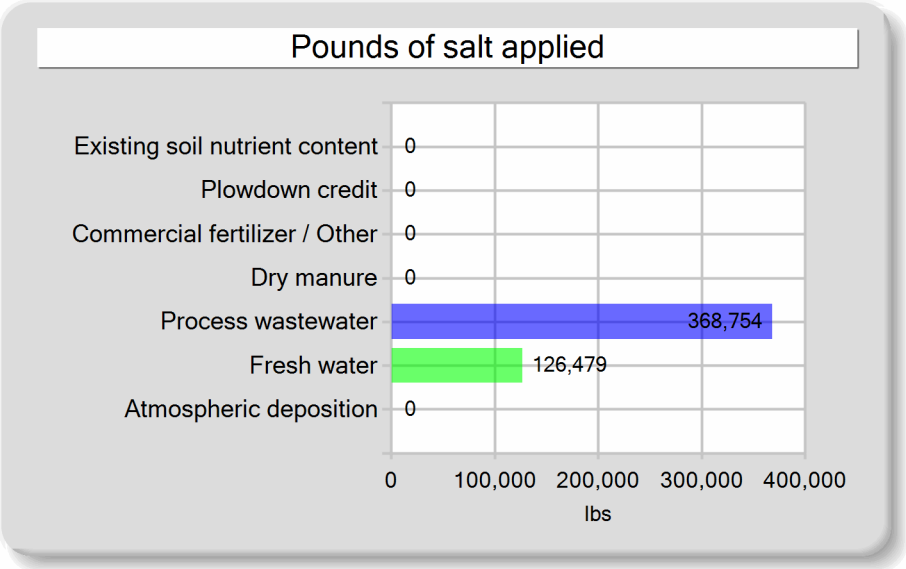


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Milk and dry cows were sold in August 2022. Only heifers remain on the facility.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael Ronald Kooyman Trust	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-24-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Kooyman & Son
6787 E Jahant Rd
Acampo, CA 95220

Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23J1797-01	IW 1	Canal Water			10/25/2023 11:00
23J1797-02	IW 3	Canal Water			10/25/2023 11:00
23J1797-03	IW 4	Canal Water			10/25/2023 11:00
23J1797-04	IW 11	Canal Water			10/25/2023 11:00
23J1797-05	DW House	Drinking Water			10/25/2023 11:00
23J1797-06	DW Milk Barn	Drinking Water			10/25/2023 11:00

Default Cooler Temperature on Receipt °C: 2.3

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

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6787 E Jahant Rd
Acampo, CA 95220

Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results

Sample: IW 1
23J1797-01 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		11/01/23 11:09	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**549**	umhos/cm	10.0	1		11/01/23 11:09	SM 2510 B		BEJ1126
Nitrate Nitrogen as NO3N	**12.7**	mg/L	0.1	1	10	10/27/23 00:34	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:09	SM 4500-H+	H	BEJ1126
pH	**7.6**	units	1.0	1		11/01/23 11:09	SM 4500-H+	H	BEJ1126
Total Filterable Solids (TDS)	**350**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:51	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**12.9**	mg/L	1.00	1		10/30/23 10:51	SM 4500-NH3 C		BEJ1110

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Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results
(Continued)

Sample: IW 3
23J1797-02 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		11/01/23 11:12	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**549**	umhos/cm	10.0	1		11/01/23 11:12	SM 2510 B		BEJ1126
Nitrate Nitrogen as NO3N	**13.0**	mg/L	0.1	1	10	10/27/23 00:55	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:12	SM 4500-H+	H	BEJ1126
pH	**7.7**	units	1.0	1		11/01/23 11:12	SM 4500-H+	H	BEJ1126
Total Filterable Solids (TDS)	**362**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:53	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**13.3**	mg/L	1.00	1		10/30/23 10:53	SM 4500-NH3 C		BEJ1110

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Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results
(Continued)

Sample: IW 4
23J1797-03 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.22**	mmhos/cm	0.01	1		11/01/23 11:13	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**218**	umhos/cm	10.0	1		11/01/23 11:13	SM 2510 B		BEJ1126
Nitrate Nitrogen as NO3N	**1.9**	mg/L	0.1	1	10	10/27/23 01:15	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:13	SM 4500-H+	H	BEJ1126
pH	**8.0**	units	1.0	1		11/01/23 11:13	SM 4500-H+	H	BEJ1126
Total Filterable Solids (TDS)	**145**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:54	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**2.06**	mg/L	1.00	1		10/30/23 10:54	SM 4500-NH3 C		BEJ1110

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6787 E Jahant Rd
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Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results
(Continued)

Sample: IW 11
23J1797-04 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.25**	mmhos/cm	0.01	1		11/01/23 11:15	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**247**	umhos/cm	10.0	1		11/01/23 11:15	SM 2510 B		BEJ1126
Nitrate Nitrogen as NO3N	**2.3**	mg/L	0.1	1	10	10/27/23 01:35	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:15	SM 4500-H+	H	BEJ1126
pH	**8.1**	units	1.0	1		11/01/23 11:15	SM 4500-H+	H	BEJ1126
Total Filterable Solids (TDS)	**160**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:55	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**2.27**	mg/L	1.00	1		10/30/23 10:55	SM 4500-NH3 C		BEJ1110

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Kooyman & Son
6787 E Jahant Rd
Acampo, CA 95220

Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results
(Continued)

Sample: DW House
23J1797-05 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		11/01/23 11:16	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**234**	umhos/cm	10.0	1		11/01/23 11:16	SM 2510 B		BEJ1126
Ammonia (as N)	*	mg/L	0.00	1		10/25/23 11:00	Field		BEJ1099
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	10/27/23 01:56	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:16	SM 4500-H+	H	BEJ1126
pH	**8.0**	units	1.0	1		11/01/23 11:16	SM 4500-H+	H	BEJ1126
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:57	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**2.30**	mg/L	1.00	1		10/30/23 10:57	SM 4500-NH3 C		BEJ1110

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Kooyman & Son
6787 E Jahant Rd
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Account# 00-0026268
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/06/2023 09:20

Sample Results
(Continued)

Sample: DW Milk Barn
23J1797-06 (Water)

Sampled: 10/25/2023 11:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.26**	mmhos/cm	0.01	1		11/01/23 11:17	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**259**	umhos/cm	10.0	1		11/01/23 11:17	SM 2510 B		BEJ1126
Ammonia (as N)	*	mg/L	0.00	1		10/25/23 11:00	Field		BEJ1099
Nitrate Nitrogen as NO3N	**2.4**	mg/L	0.1	1	10	10/27/23 02:16	EPA 300.0		BEJ1063
Temperature	**25.0**	units	0.0	1		11/01/23 11:17	SM 4500-H+	H	BEJ1126
pH	**7.8**	units	1.0	1		11/01/23 11:17	SM 4500-H+	H	BEJ1126
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 11:06	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**2.42**	mg/L	1.00	1		10/30/23 11:06	SM 4500-NH3 C		BEJ1110

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

10/26/23 16:45 23J1797

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: Acct # ~~8~~ 26268 Cons # 13 19

No. Samples: 6 No of Bottles: 12

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Kooyman & Son
Address: 6787 E Jahant Rd
City: Acampo State: CA Zip: 95220
Telephone: 209 367-4426 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1 +TC
DWW1 ++C

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples		Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW	1	10/25/23	started 11 AM	2.3	
2	↓	3	↓	↓	2.2	
3	↓	4	↓	↓	2.9	
4	↓	11	↓	↓	1.8	
5	DW	House	↓	↓	Ø	
6	↓	Milk Barn	↓	↓	0.1	
7						
8						
9						
10						

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JCF	DLV	10-26-23 10:00	10-26-23 4:30
Third				
Fourth	[illegible]	DLV	10/26 1445	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No